CONFIDENTIAL **MOCK EXAM 2026**

GENERIC UNIVERSITY FINAL EXAMINATION

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

PART A (80 MARKS)

QUESTION 1

a) Using Caesar Cipher technique, decrypt the following ciphertext with a shift of 7 (replace each letter with the one 7 positions to the left).
KLCPJLYLABYULKHUKDPWLKVUL

(2 marks)

b) Describe substitution ciphers by explaining what operation they perform on plaintext, what serves as the key, which security principle they relate to, and how larger blocks and varying substitutions are used to overcome their weaknesses. Include six points.

(6 marks)

QUESTION 2

a) Explain how a chain of trust is established in the Trusted Computing architecture, including the role played by the TPM and the roots of trust.

(2 marks)

b) Refer to your answer in (a). Explain six components and characteristics of the NGSCB/Nexus architecture, covering the Nexus kernel, the Nexus Computing Agent, and how these relate to the TPM and the platform's trusted mode.

(6 marks)

QUESTION 3

a) List TWO (2) reasons why software is insecure, with particular reference to buffer overflows.

(2 marks)

b) Describe six (6) aspects of the security mechanisms found in Windows systems, covering the overall mechanism set, access control, subjects and credentials, access rights, directory services, and the risks arising from complexity.

(6 marks)

QUESTION 4

a) State TWO (2) formal security models that are listed as key security models, and for each, state the specific security concern it addresses.

(2 marks)

b) Explain the access control structures used to implement access policies, covering the access control matrix, access control lists and capabilities, and state the requirements such structures must satisfy.

(6 marks)

QUESTION 5

a) Explain two characteristics of Trojan horse malware, including how it disguises itself and what harm it can cause once active.

(2 marks)

b) Discuss six ways in which malware attacks and propagates through a computer system, referring to the techniques malware uses to spread, hide, and affect the infected host.

(6 marks)

QUESTION 6

a) State TWO (2) impacts of a security attack on the confidentiality, integrity or availability of a system's assets.

(2 marks)

b) Discuss the Samy worm as an example of a cross-site scripting (XSS) attack, covering how the malicious script was injected, how it spread between users, and the scale of the resulting infection.

(6 marks)

QUESTION 7

a) State TWO (2) differences between morals and laws as described in the lecture on ethics and computer security.

(2 marks)

b) The ACM Code of Ethics and Professional Conduct guides computing professionals. State SIX (6) obligations that the ACM code places on a computing professional in their conduct and ethical practice.

(6 marks)

QUESTION 8

a) Explain two ways in which an intrusion detection system contributes to detecting and defending against attacks on a computer system.

(2 marks)

b) State six (6) points about threat classes and the countermeasures used against them, covering how threats differ from vulnerabilities and attacks, how misuse-based detection recognises known attack patterns, and how preventive mechanisms such as firewalls, sandboxing and input sanitisation reduce the risk from malicious activity.

(6 marks)

QUESTION 9

a) Explain where network-based IDS sensors should be placed in a network and why their placement matters.

(2 marks)

b) Explain six preventive network security mechanisms and services that an organization can deploy to protect its network, describing what each mechanism does and how it contributes to securing network access and communication.

(6 marks)

QUESTION 10

a) List the TWO (2) stages of the biometric authentication process.

(2 marks)

b) Compare the two categories of biometric applications available today by identifying the biometric technologies that belong to each category, and then discuss three disadvantages associated with fingerprint biometrics.

(6 marks)

PART B (20 MARKS)

QUESTION 1

a) State the role of a reference monitor in authorization and outline the four goals of access control. Then state the specific concern that the Chinese Wall policy is designed to address and explain why its permissions change dynamically.

(5 marks)

b) Explain the Chinese Wall model by stating the conflict it is designed to resolve, its classification as a policy type, the condition under which a subject may be granted access to an object, and the two conditions that must hold before write access is permitted.

(5 marks)

QUESTION 2

a) Discuss FIVE (5) password security controls that an administrator can apply to protect user accounts on a Linux system, and explain how each control reduces the risk of unauthorized access.

(5 marks)

b) Explain the design criteria that Kerberos aims to satisfy and the benefits that single sign-on solutions such as Kerberos bring to large distributed environments.

(5 marks)

END OF QUESTION PAPER